

Problem 13.6.1

Show that

a) $\sinh z = \sinh x \cos y + i \cosh x \sin y$

b) $\cosh z = \cosh x \cos y + i \sinh x \sin y$

Solution

a)

$$\begin{aligned}
 \sinh z &= \frac{1}{2}(e^z - e^{-z}) \\
 &= \frac{1}{2}(e^{x+iy} - e^{-x-iy}) \\
 &= \frac{1}{2}(e^x e^{iy} - e^{-x} e^{-iy}) \\
 &= \frac{1}{2}(e^x (\cos y + i \sin y) - e^{-x} (\cos y - i \sin y)) \\
 &= \frac{1}{2}(e^x \cos y + i e^x \sin y - e^{-x} \cos y + i e^{-x} \sin y) \\
 &= \frac{1}{2}((e^x - e^{-x}) \cos y + i(e^x + e^{-x}) \sin y) \\
 &= \boxed{\sinh x \cos y + i \cosh x \sin y}
 \end{aligned}$$

b)

$$\begin{aligned}
 \cosh z &= \frac{1}{2}(e^z + e^{-z}) \\
 &= \frac{1}{2}(e^{x+iy} + e^{-x-iy}) \\
 &= \frac{1}{2}(e^x e^{iy} + e^{-x} e^{-iy}) \\
 &= \frac{1}{2}(e^x (\cos y + i \sin y) + e^{-x} (\cos y - i \sin y)) \\
 &= \frac{1}{2}(e^x \cos y + i e^x \sin y + e^{-x} \cos y - i e^{-x} \sin y) \\
 &= \frac{1}{2}((e^x + e^{-x}) \cos y + i(e^x - e^{-x}) \sin y) \\
 &= \boxed{\cosh x \cos y + i \sinh x \sin y}
 \end{aligned}$$

Problem 13.6.3

Show that

a) $\cosh^2 z - \sinh^2 z = 1$

b) $\cosh^2 z + \sinh^2 z = \cosh 2z$

Solution

a)

$$\begin{aligned}
 \cosh^2 z - \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 - \frac{1}{4}(e^z - e^{-z})^2 \\
 &= \frac{1}{4}[(e^{2z} + 2 + e^{-2z}) - (e^{2z} - 2 + e^{-2z})] \\
 &= \boxed{1}
 \end{aligned}$$

b)

$$\begin{aligned}
 \cosh^2 z + \sinh^2 z &= \frac{1}{4}(e^z + e^{-z})^2 + \frac{1}{4}(e^z - e^{-z})^2 \\
 &= \frac{1}{4}[(e^{2z} + 2 + e^{-2z}) + (e^{2z} - 2 + e^{-2z})] \\
 &= \frac{1}{4}(2e^{2z} + 2e^{-2z}) \\
 &= \frac{1}{2}(e^{2z} + e^{-2z}) \\
 &= \boxed{\cosh 2z}
 \end{aligned}$$

Problem 13.6.7Find, in the form $u + iv$

a) $\cos i$

b) $\sin i$

Solution

a)

$$\cos i = \boxed{\cosh 1}$$

b)

$$\sin i = \boxed{i \sinh 1}$$

Problem 13.6.13

Using the definitions, prove

a) $\cos z$ is even, $\cos(-z) = \cos z$

b) $\sin z$ is odd, $\sin(-z) = -\sin z$

Solution

a)

$$\begin{aligned}
 \cos(-z) &= \frac{1}{2}(e^{i \cdot (-z)} + e^{-i \cdot (-z)}) \\
 &= \frac{1}{2}(e^{-iz} + e^{iz}) \\
 &= \boxed{\cos z}
 \end{aligned}$$

b)

$$\begin{aligned}
 \sin(-z) &= \frac{1}{2i} \left(e^{i \cdot (-z)} - e^{-i \cdot (-z)} \right) \\
 &= \frac{1}{2i} (e^{-iz} - e^{iz}) \\
 &= \boxed{-\sin z}
 \end{aligned}$$

Problem 13.6.16

Find all solutions

$$\sin z = 100$$

Solution

$$\sin z = \frac{1}{2i} (e^{iz} - e^{-iz}) = 100$$

Let $w = e^{iz}$. Then $w - w^{-1} = 100i$

$$w^2 - 200iw - 1 = 0 \quad \implies \quad w = i(100 \pm \sqrt{9999})$$

Thus $e^{iz} = i(100 \pm \sqrt{9999})$, so

$$\begin{aligned}
 iz &= \ln(i(100 \pm \sqrt{9999})) \\
 &= \ln|i(100 \pm \sqrt{9999})| + i \operatorname{Arg}(i(100 \pm \sqrt{9999})) + i2\pi n \\
 &= \ln(100 \pm \sqrt{9999}) + i\frac{\pi}{2} + i2\pi n \\
 &= \pm \ln(100 + \sqrt{9999}) + i\pi \left(2n + \frac{1}{2} \right)
 \end{aligned}$$

Dividing by i ,

$$z = \pi \left(2n + \frac{1}{2} \right) \pm \ln(100 + \sqrt{9999}) \quad n \in \mathbb{Z}$$

Problem 13.7.5

Find

$$\operatorname{Ln}(-11)$$

Solution

$$\begin{aligned}
 \operatorname{Ln}(-11) &= \ln|-11| + i \operatorname{Arg}(-11) \\
 &= \boxed{\ln 11 + i\pi}
 \end{aligned}$$

Problem 13.7.7

Find

$$\operatorname{Ln}(4 - 4i)$$

Solution

$$\begin{aligned}\operatorname{Ln}(4 - 4i) &= \ln |4 - 4i| + i \operatorname{Arg}(4 - 4i) \\ &= \ln \sqrt{4^2 + 4^2} + i \tan^{-1} \left(\frac{-4}{4} \right) \\ &= \ln \sqrt{32} + i \tan^{-1}(-1) \\ &= \boxed{\frac{1}{2} \ln 32 - i \frac{\pi}{4}}\end{aligned}$$

Problem 13.7.8

Find

$$\operatorname{Ln}(1 \pm i)$$

Solution

$$\begin{aligned}\operatorname{Ln}(1 \pm i) &= \ln |1 \pm i| + i \operatorname{Arg}(1 \pm i) \\ &= \ln \sqrt{2} + i \tan^{-1} \left(\frac{\pm 1}{1} \right) \\ &= \ln \sqrt{2} + i \tan^{-1}(\pm 1) \\ &= \boxed{\frac{1}{2} \ln 2 \pm i \frac{\pi}{4}}\end{aligned}$$

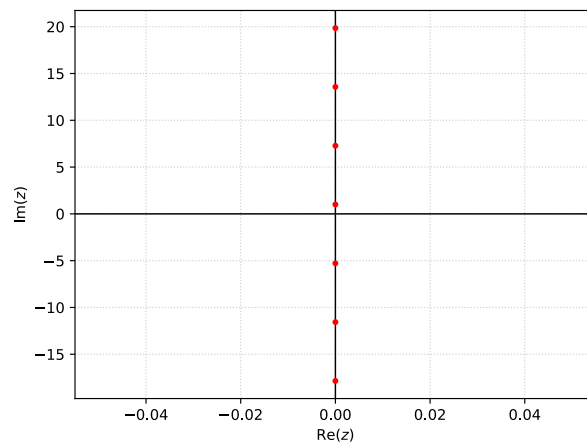
Problem 13.7.15

Find all values and graph some in the complex plane

$$\ln(e^i)$$

Solution

$$\begin{aligned}\ln(e^i) &= \ln |e^i| + i \operatorname{Arg}(e^i) + i2\pi n \\ &= \ln 1 + i + i2\pi n \\ &= \boxed{i(1 + 2\pi n) \quad n \in \mathbb{Z}}\end{aligned}$$

Figure 1: Solutions of $\ln(e^i)$ **Problem 13.7.19**Solve for z

$$\ln z = 4 - 3i$$

Solution

$$\begin{aligned}
 e^{\ln z} &= e^{4-3i} \\
 z &= e^{4-3i} \\
 &= e^4 (\cos(-3) + i \sin(-3)) \\
 &= \boxed{e^4 (\cos 3 - i \sin 3)}
 \end{aligned}$$

Problem 13.7.23

Find the principal value.

$$(1+i)^{1-i}$$

Solution

$$\begin{aligned}
(1+i)^{1-i} &= \exp((1-i) \operatorname{Ln}(1+i)) \\
&= \exp\left((1-i) \left(\frac{1}{2} \ln 2 + i \frac{\pi}{4}\right)\right) && \text{(from 13.7.8)} \\
&= \exp\left((1-i) \left(\frac{1}{2} \ln 2 + i \frac{\pi}{4}\right)\right) \\
&= \exp\left(\frac{1}{2} \ln 2 + i \frac{\pi}{4} - i \frac{1}{2} \ln 2 + \frac{\pi}{4}\right) \\
&= \exp\left(\frac{\pi}{4} + \frac{1}{2} \ln 2\right) \exp\left(i \left(\frac{\pi}{4} - \frac{1}{2} \ln 2\right)\right) \\
&= \boxed{\exp\left(\frac{\pi}{4} + \frac{1}{2} \ln 2\right) \left[\cos\left(\frac{\pi}{4} - \frac{1}{2} \ln 2\right) + i \sin\left(\frac{\pi}{4} - \frac{1}{2} \ln 2\right)\right]}
\end{aligned}$$

Problem 14.1.1

Find and sketch the path

$$z(t) = \left(1 + \frac{1}{2}i\right)t \quad (2 \leq t \leq 5)$$

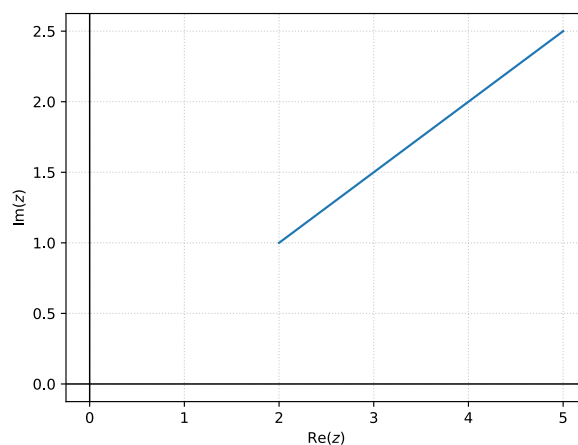
Solution

Figure 2: $z(t)$

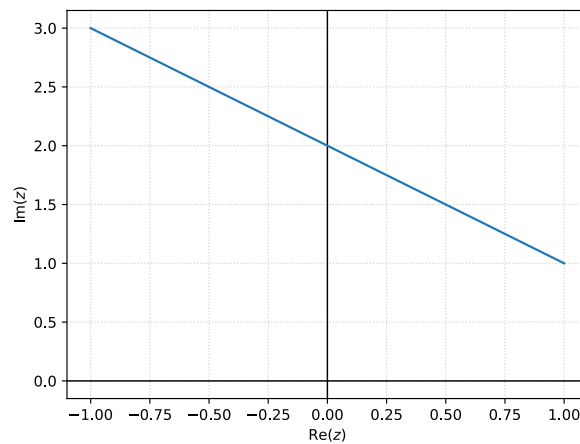
Problem 14.1.11

Find a parametric representation and sketch the path

Segment from $(-1, 1)$ to $(1, 3)$

Solution

$$z(t) = (t-1) + i(t+1) \quad t \in [0, 2]$$

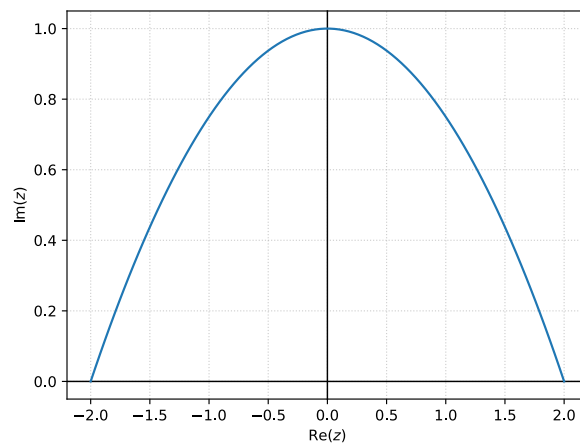
Figure 3: $z(t)$ **Problem 14.1.19**

Find a parametric representation and sketch the path

$$\text{Parabola } y = 1 - \frac{1}{4}x^2 \quad (-2 \leq x \leq 2)$$

Solution

$$z(t) = t + i \left(1 - \frac{1}{4}t^2 \right) \quad t \in [-2, 2]$$

Figure 4: $z(t)$ **Problem 14.1.21**

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Re} z \, dz, \quad C \text{ the shortest path from } 1 + i \text{ to } 3 + 3i$$

Solution

$$f(z) = \operatorname{Re} z = u(x, y) + iv(x, y)$$

$$\begin{aligned} u &= x & v &= 0 \\ u_x &= 1 & v_x &= 0 \\ u_y &= 0 & v_y &= 0 \end{aligned}$$

$u_x \neq v_y$ therefore f is not analytic and the first method does not apply.

$$\int_C f(z) \, dz = \int_a^b f[z(t)] \dot{z}(t) \, dt$$

$$\begin{aligned} z(t) &= t + it & t &\in [1, 3] \\ \dot{z}(t) &= 1 + i \end{aligned}$$

$$\begin{aligned} \int_C f(z) \, dz &= \int_1^3 f[z(t)] \dot{z}(t) \, dt \\ &= \int_1^3 t \cdot (1 + i) \, dt \\ &= (1 + i) \frac{1}{2} t^2 \Big|_1^3 \\ &= \boxed{4 + 4i} \end{aligned}$$

Problem 14.1.25

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C z \exp(z^2) \, dz, \quad C \text{ from } 1 \text{ along the axes to } i$$

Solution

$$\begin{aligned} f(z) &= z \exp(z^2) \\ f'(z) &= \exp(z^2) + 2z^2 \exp(z^2) \end{aligned}$$

$f'(z)$ is defined everywhere and is therefore entire. The first method applies.

$$\int_C z \exp(z^2) \, dz = \int_1^i z \exp(z^2) \, dz$$

Substituting $u = z^2$, $du = 2z \, dz$,

$$\begin{aligned}
\int_1^i z \exp(z^2) dz &= \frac{1}{2} \int_1^{-1} \exp(u) du \\
&= \frac{1}{2} \exp(u) \Big|_1^{-1} \\
&= \frac{1}{2} (\exp(-1) - \exp(1)) \\
&= \boxed{-\sinh(1)}
\end{aligned}$$

Problem 14.1.29

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Im} z^2 dz \quad \text{counterclockwise around the triangle with vertices } 0, 1, i$$

Solution

$$\begin{aligned}
f(z) = u(x, y) + iv(x, y) &= \operatorname{Im} z^2 \\
&= \operatorname{Im} x^2 - y^2 + 2ixy \\
&= 2xy
\end{aligned}$$

$$\begin{aligned}
u &= 0 & v &= 2xy \\
u_x &= 0 & v_x &= 2y \\
u_y &= 0 & v_y &= 2x
\end{aligned}$$

$u_x \neq v_y$ and $u_y \neq -v_x$ therefore f is not analytic and the first method does not apply.

$$\int_C f(z) dz = \int_a^b f[z(t)] \dot{z}(t) dt$$

$$\begin{aligned}
z_1(t) &= t & \dot{z}_1(t) &= 1 & t &\in [0, 1] \\
z_2(t) &= (1-t) + it & \dot{z}_2(t) &= -1 + i & t &\in [0, 1] \\
z_3(t) &= i(1-t) & \dot{z}_3(t) &= -i & t &\in [0, 1]
\end{aligned}$$

$$\begin{aligned}
\int_C \operatorname{Im} z^2 dz &= \int_0^1 f(z_1) \dot{z}_1 dt + \int_0^1 f(z_2) \dot{z}_2 dt + \int_0^1 f(z_3) \dot{z}_3 dt \\
&= \int_0^1 \operatorname{Im}(t^2) dt + \int_0^1 \operatorname{Im}((1-t) + it)^2 (-1 + i) dt + \int_0^1 \operatorname{Im}(i(1-t))^2 (-i) dt \\
&= \int_0^1 2(t)(0) dt + (-1 + i) \int_0^1 2(1-t)(t) dt - i \int_0^1 2(0)(1-t) dt \\
&= 2(-1 + i) \int_0^1 (t - t^2) dt \\
&= 2(-1 + i) \left[\frac{t^2}{2} - \frac{t^3}{3} \right]_0^1 \\
&= \boxed{-\frac{1}{3} + \frac{i}{3}}
\end{aligned}$$